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Review

Fast and furious: The rapid turnover of microRNAs in plants

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Abstract Plant miRNAs exhibit a dynamic and complex evolutionary landscape. Despite their rapid turnover, miRNAs play crucial roles in regulating key biological processes, underscoring their functional significance even when evolutionarily transient. This review explores the phenomenon of miRNA turnover through a comprehensive survey of miRNA conservation across land plants. We discuss how *de novo* miRNAs overcome challenges such as transcriptional activation, structural requirements for biogenesis, and functional integration into gene regulatory networks. Furthermore, we review the mechanisms of miRNA origination, including inverted duplication of target genes, derivation from retrotransposons or DNA transposons, spontaneous evolution, and a newly proposed mechanism through template switching mutations. Duplication of existing miRNAs plays a significant role in miRNA family expansion, driving the functional diversification and strengthening regulatory networks. While the phenomenon of miRNA loss has been preliminarily explored, its mechanisms remain insufficiently understood. To address this, we proposed three detailed steps to advance research into miRNA loss. This review provides an integrated perspective on the gain, expansion, and loss of plant miRNAs, offering insights into their evolutionary and biological significance.

Key words: expansion, gain, loss, microRNAs, plants.

1 Introduction

MicroRNAs (miRNAs) are a unique class of non-coding RNAs that play crucial roles in the regulation of gene expression. Most miRNAs function by base-pairing with target mRNA transcripts, leading to either translational repression, a mechanism predominantly observed in animals, or mRNA cleavage, which is more common in plants (Voinnet, 2009; Bartel, 2018; Shang et al., 2023). Additionally, some miRNAs can regulate their targets through other epigenetic modifications, such as DNA methylation (Wu et al., 2010; Creasey et al., 2014). Since the discovery of the first batch of plant miRNAs over two decades ago (Reinhart et al., 2002), miRNAs have been profoundly recognized for their functions in a broad spectrum of biological processes. In plants, miRNAs are integral to regulation of growth and development, influencing key processes such as the timing of flowering, seed germination, and organ morphogenesis (Jones-Rhoades et al., 2006; Wang et al., 2009; Song et al., 2019; Jiang et al., 2021). Additionally, miRNAs play pivotal roles in mediating responses to both biotic and abiotic stresses, as well as facilitating interspecies communication (Sunkar et al., 2012; Pagano et al., 2021; Chen & Rechavi, 2022). This comprehensive understanding of miRNA functions highlights their potential applications in agriculture for improving crop productivity and resilience.

The advent of next-generation sequencing (NGS) technologies marked a pivotal milestone in miRNA research, ushering in the era of high-throughput miRNA identification. Simultaneously, many bioinformatics tools, such as the series of miRDP (Yang & Li, 2011; Kuang et al., 2019) and sRNAminer (Li et al., 2024), have been developed to identify and annotate plant miRNAs by providing more precise and efficient analyses. With the accumulation of miRNA entries and a deeper understanding of miRNA biology, machine learning-based annotation tools, such as iwa-miRNA (Zhang et al., 2022b), have further advanced progress by directly predicting miRNAs from genomic sequences. These advancements facilitate large-scale miRNA identification across diverse plant species, providing foundational data for miRNA evolution research.

The evolution of plant miRNAs is a dynamic process characterized by rapid turnover, meaning frequent gain and loss events occurring over relatively short evolutionary timescales, especially during speciation. This turnover is considered "rapid" when researchers made the comparison of miRNAs between phylogenetically close species, such as the comparison between *Arabidopsis thaliana* and *Arabidopsis lyrate* (Fahlgren et al., 2010; Ma et al., 2010; Guo et al., 2022b). Only a few miRNA families are highly conserved across land plants, the majority are lineage-specific or species-specific, underscoring the high turnover within the miRNA repertoire. The relatively limited number of miRNAs within an individual plant genome, typically ranging from tens to hundreds, suggests that miRNAs have undergone swift gains and losses throughout the evolution of plant species (Cui et al., 2017; Guo et al., 2022a, 2022b).

The evolution of miRNAs is closely connected to other types of small RNAs (sRNAs). In plants, sRNAs are predominantly classified into two main types, miRNAs and short interfering RNAs (siRNAs) (Vaucheret & Voinnet, 2023). siRNAs are derived from perfect, often extended hairpins and are primarily processed by DCL2, DCL3, and/or DCL4 into siRNA populations (Henderson et al., 2006; Dunoyer et al., 2007). Most miRNAs are generated from single-stranded RNAs that form short, bulged hairpin structures, which are typically processed by DICER-LIKE1 (DCL1) to produce a single, imperfectly paired miRNA/miRNA* duplex (Voinnet, 2009; Rosatti et al., 2024). Additionally, a subset of miRNAs, referred to as young miRNAs, are produced from longer, imperfect hairpins that are processed by DCL4 into multiple duplexes (Rajagopalan et al., 2006).

In some cases, DCL1 processes a single precursor to produce multiple miRNA/miRNA*, referred to as sequential *MIRNAs*, such as *MIR159a*, *MIR169m* and *MIR319a/b* (Zhang et al., 2010). A recent study analyzed the interacting energies among these duplexes derived from the same precursor and found that the primary expressed duplex was more stable than others (Rosatti et al., 2024). Notably, the energy differences in conserved sequential miRNAs are more pronounced than those in young miRNAs. For example, the evolutionarily young *MIR163* generates two miRNA/miRNA* duplexes with an energy difference of <6 kcal/mol, compared to the 11 kcal/mol average difference in conserved sequential *MIRNAs*. Consequently, the study proposed that evolutionarily young sequential *MIRNAs* produce multiple duplexes with similar energy and expression level, while ancient sequential *MIRNAs* predominantly generate a single miRNA/miRNA* due to the additional mutations in other duplexes.

This review aims to provide a comprehensive overview of the rapid turnover of miRNAs, focusing on the mechanisms driving the gain, expansion, and loss of miRNAs in plants. Notably, we highlight the novelty and significance of miRNA loss in the further study.

2 Rapid Turnover of miRNAs

2.1 Conserved, lineage-specific, and species-specific miRNAs

The rapid gain and loss of miRNAs is a well-recognized phenomenon in evolutionary biology (Meunier et al., 2013; Dexheimer & Cochella, 2020). Although related studies in animals remain controversial (Tarver et al., 2018), the swift turnover of miRNAs in plants is well-documented, leading to varying degrees of miRNA conservation across plants. Notably, there is lack of convincing evidence for miRNA sequence homology between land plants and other groups such as algae, amoebae, and animals (Avesson et al., 2012; Tarver et al., 2015; Cock et al., 2017; Moran et al., 2017). Therefore, this review focuses on the conservation of miRNAs within land plants.

Using annotated miRNAs from the PmiREN database (https://www.pmiren.com) (Guo et al., 2020, 2022c), we conducted a comprehensive survey of miRNA conservation across 109 land plants (Fig. 1). There are seven miRNA families, miR156, miR160, miR166, miR171, miR319, miR390, and miR408, demonstrate a highest degree of conservation in land plants over hundreds of millions of years (Axtell & Bartel, 2005; Cuperus et al., 2011; Guo et al., 2020). The widespread presence of these conserved miRNAs across nearly all sampled species highlights their fundamental roles in plant biology, particularly in regulating developmental processes (Jones-Rhoades, 2012; Liu et al., 2018; Millar, 2020). This conservation likely provides an evolutionary advantage, contributing to the successful colonization of terrestrial habitats by plants. For instance, miR156 triggers the transition from juvenile to adult phases by targeting SQUAMOSA promoter binding protein-like (SPL) transcription factors, which promote adult traits and flowering (Wang et al., 2009; Jiao et al., 2010). MiR160 targets auxin response factors (ARFs), which are crucial regulators of plant growth and development (Mallory et al., 2005; Wang et al., 2005). MiR166 regulates class III homeodomain leucine-zipper (HD-ZIP III) proteins, essential for the meristem and vascular tissue development (Ko et al., 2006; Sakaguchi & Watanabe, 2012). Additionally, miR408 involves in copper allocation to the chloroplast, contributing to biomass yield, drought resistance, and leaf senescence (Pan et al., 2018; Hao et al., 2022; Guo et al., 2023; Yang et al., 2024).

While the aforementioned miRNAs are highly conserved across land plants, the majority of miRNAs are not as broadly conserved and are instead lineage-specific or species-specific. Among these, at least three families, miR159, miR396, and miR172, are conserved within vascular plants (Fig. 1). Other miRNA families, including miR164, miR167, miR168, miR169, miR393, miR394, miR395, miR397, miR162, and miR170, are conserved in seed plants. Additionally, miR528 and miR1432 are conserved in monocots, while miR1511, miR1515, and miR1507 are specific to eudicots. Some miRNA families have undergone lineage-specific loss, such as the loss of miR157 in monocots and the loss of miR529 in core eudicots (Morea et al., 2016). These patterns of conservation and loss highlight the dynamic evolution of miRNAs.

The most stringent definition of species-specific miRNAs refers to those unique to a single plant species. However, due to current limitations in miRNA annotation across all species, many studies define miRNAs as species-specific if they are detected in only one species and are absent from other species with annotated miRNAs (Li et al., 2012; Jaiswal et al., 2019; Guo et al., 2024). In a recent study that comprehensively analyzed miRNAs across 81 plant species, ranging from chlorophytes to angiosperms, researchers identified 8,048 species-specific miRNAs from 5,499 families, representing over 61.2% of the miRNA families in the examined species (Guo et al., 2022a). The analysis revealed a significant bias in the genomic coordinates, with species-specific miRNAs more frequently located near or within genes. Functional analyses indicated that these miRNAs preferentially regulate diverse metabolic processes, potentially contributing to the unique traits and adaptations of plants to specific environments. For example, a rice-specific miRNA, miRN2285, which originates through miniature inverted-repeat transposable elements (MITEs), contributes to cold resistance in rice varieties cultivated at high latitudes (Guo et al., 2022b).

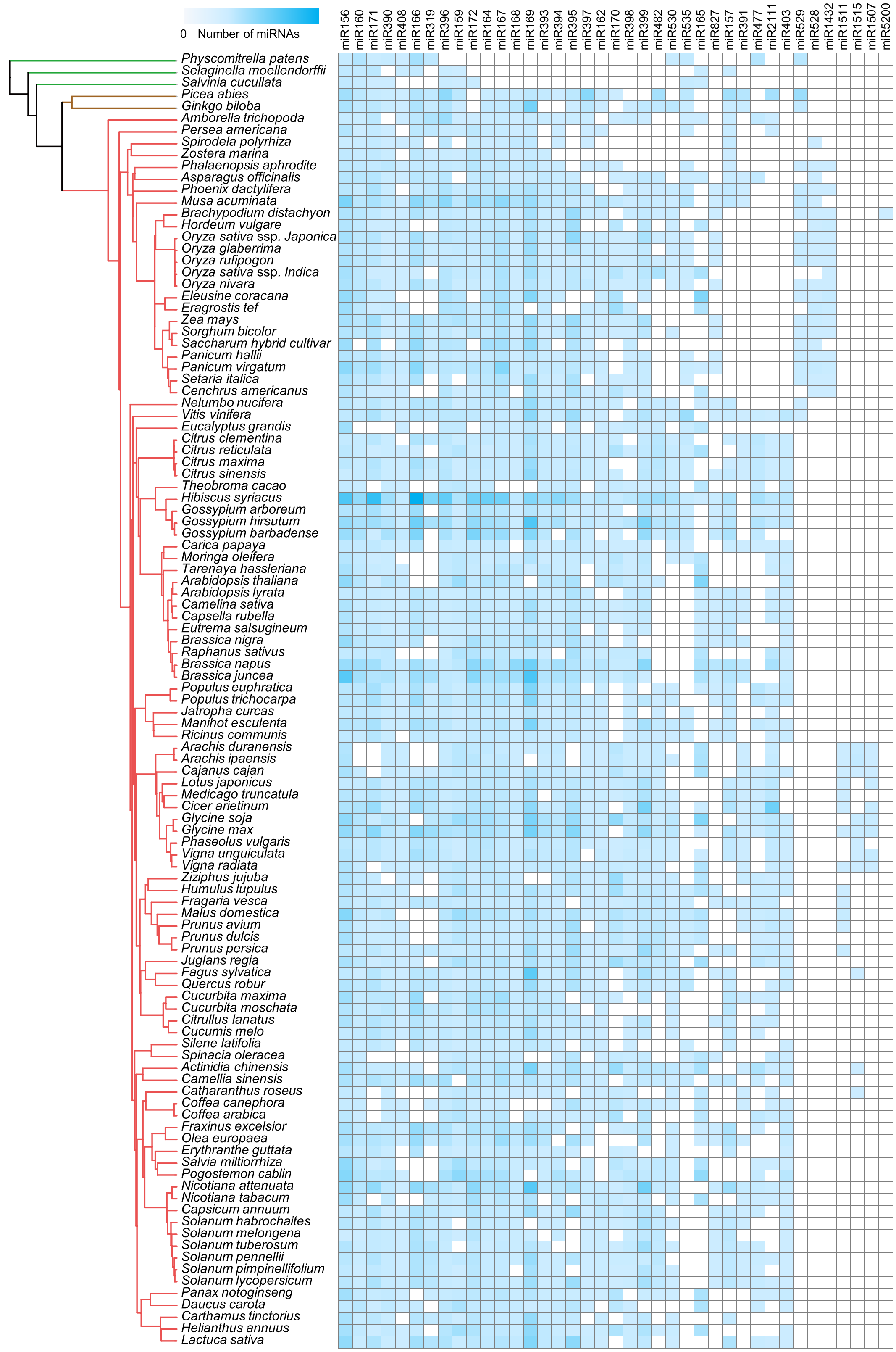

Fig. 1. Conserved and specific miRNA families in land plants. miRNAs from 109 species were retrieved from the PmiREN database. The presence of a miRNA family (columns) in a species (rows) is indicated by blue shading. The gradient from light to dark blue represents the number of miRNA paralogs within each family from low to high. Most lineage-specific and species-specific miRNA families were omitted due to space limitation. The species tree was retrieved from the TimeTree website (Kumar et al., 2022).

2.2 The paradox of evolutionarily transient and functional significance

Some studies, particularly in animals, suggest that the emergence of new miRNAs could be detrimental to the organism. As a result, the expression levels of these miRNAs are often kept low to avoid being eliminated (Begun et al., 2007; Lu et al., 2008; Meunier et al., 2013; Patel & Capra, 2017). Furthermore, research on gene regulatory networks reveals that only a small fraction of new miRNAs successfully integrate into networks to perform functional roles (Chen & Rajewsky, 2007; Wang et al., 2023).

Conversely, research primarily in plants argues that the generation of new miRNAs plays a positive role to the organism. Investigations into the target genes of new miRNAs show that most of these miRNAs have specific targets, thereby enriching gene regulatory networks and enhancing adaptability to environmental changes (Gao et al., 2021; Pagano et al., 2021; Zhang et al., 2022a). For instance, in rice, long miRNAs originating from MITEs exhibit dynamic expression changes under stress conditions such as pathogen, drought, or extreme temperatures, and target genes are enriched in stress response pathways (Huang et al., 2022). Analyses in angiosperms suggest that new miRNAs are more likely to regulate genes involved in stress responses and secondary metabolism, potentially enhancing the adaptability of plants to complex environments (Guo et al., 2022a, 2022b).

Moreover, specific miRNAs have been shown to play significant roles in various plant life processes. For example, miR528, specific in monocots, regulates multiple copper-containing proteins involved in lignin polymerization, viral resistance, and peel browning (Zhu et al., 2020). A miRNA specific in *Arabidopsis thaliana*, miR775, controls cell and organ size by regulating pectin content and cell wall elasticity (Zhang et al., 2021). Three newly identified miRNAs in the *Ilex latifolia* may be involved in determining the deciduous or evergreen traits among closely related species (Guo et al., 2024). Also, studies in animals suggest that new miRNA families are associated with body-plan innovations and an increase in overall morphological complexity (Sempere et al., 2006; Deline et al., 2018; Zolotarov et al., 2022; Langschied et al., 2023). These findings indicate that even transient miRNAs could have significant functional roles.

Therefore, a fascinating paradox is obviously present. On one hand, miRNAs exhibit a high turnover rate, with new miRNAs frequently emerging while others being lost over relatively short evolutionary timescales. On the other hand, even lineage-specific and species-specific miRNAs often play crucial roles in regulating key biological processes. This duality raises intriguing questions about how miRNAs can be both evolutionarily transient and functionally significant.

Here, we propose a plausible explanation for this paradox. Conserved miRNAs often regulate fundamental processes that are vital for survival. In contrast, lineage-specific and species-specific miRNAs are more transient, providing the flexibility for plants to adapt to new environmental challenges. This balance allows plants to respond rapidly to new selective pressures while preserving the core regulatory networks necessary for their survival, highlighting the delicate equilibrium between innovation and conservation in evolutionary processes. We believe that as more specific miRNAs are identified and their biological functions are revealed, this explanation will be validated and refined.

3 Gain, Expansion, and Loss of miRNAs

3.1 Origination via *de novo* mechanism

The *de novo* origination of *MIR* genes is a complex process influenced by several stringent factors, which must be overcome for a new miRNA to emerge successfully.

First, a new *MIR* gene must achieve successful transcription. This process typically involves acquiring specific promoter elements that ensure the *MIR* gene is expressed in the appropriate spatial and temporal patterns. The distinct genomic positioning of *MIR* genes in plants and animals further suggests the distinct mechanisms required for transcriptional activation. In animals, a substantial proportion of *MIR* genes are located within the introns of protein-coding genes, allowing them to be co-transcribed with their host genes through shared promoters (Kim, 2007; Axtell et al., 2011). This intronic location facilitates the efficient integration of new miRNAs into existing transcriptional frameworks. In contrast, *MIR* genes in plants are less commonly found within introns and are more frequently located in intergenic regions (Michael, 2011; Knop et al., 2016). Consequently, the transcriptional activation of new *MIR* genes in plants often requires the acquisition of independent promoter elements or relies on the utilization of nearby promoters with bidirectional activity or read-through transcription from upstream genes.

Second, the structural demands for miRNA biogenesis are notably stringent. A precursor RNA must adopt an imperfect hairpin structure that is recognized and cleaved by DCL1 proteins to process into a functional miRNA. This structural constraint inherently limits the pool of potential *MIR* genes, as it requires precise pairing between consecutive bases on the opposite sides of the stem. Moreover, the variation in the processing of stem-loop secondary structures between plant and animal miRNAs add further complexity to this process. In animals, the stem-loop structures that produce miRNA/mRNA* duplexes are relatively uniform in size, with a typical separation of about 11 nucleotides between the base of the stem and the Drosha cleavage site (Lim et al., 2003; Han et al., 2006; Axtell, 2008; Kim et al., 2021). In contrast, plants exhibit significant variability in the sizes of stem-loops across different miRNAs, with some even reaching up to 2100 nucleotides, as reported recently (Rajagopalan et al., 2006; Axtell, 2008; Arribas-Hernández, 2024; Wang et al., 2024). Additionally, plant miRNAs often undergo suboptimal processing, a situation that can be improved by correcting the mismatches in the miRNA/miRNA* duplex (Rosatti et al., 2024). These disparities imply the distinct evolutionary pressures during the emergence of new *MIR* genes in plants and animals.

Third, integrating new miRNAs into the existing gene regulatory network poses a considerable challenge. This involves not only binding with AGO proteins but also recognizing target mRNAs with sufficient sequence complementarity to exert regulatory effects. While young *MIR* genes may acquire expression and processing capabilities, they do not always develop regulatory targets that offer a positive selection advantage (Chen & Rajewsky, 2007).

Therefore, the *de novo* origination of *MIR* genes is a complex process primarily influenced by three critical factors, transcriptional activation, structural requirements, and functional integration into gene regulatory networks. Successfully overcoming these challenges is crucial for the emergence of new *MIR* genes. These processes differ significantly between plants and animals, likely due to their distinct evolutionary trajectories. To elucidate how these factors contribute to miRNA formation, researchers have proposed several hypotheses, including inverted duplication of target genes, origination from transposable elements (TEs), spontaneous evolution, as well as a newly identified mechanism in animals by template switching mutations. The following subsections will explore these mechanisms in detail. It is noteworthy that these mechanisms are not mutually exclusive. Our observation indicates that some miRNAs exhibit sequence similarities to both protein-coding genes and transposons (Guo et al., 2022b), implying that certain miRNAs potentially arise through a combination of these processes.

3.1.1 Inverted duplication of target genes

The *de novo* origin of miRNAs through inverted duplication of target genes arose from observations of extended sequence similarities between *MIR* genes and their corresponding target genes (Allen et al., 2004; Voinnet, 2004). This model delineates a stepwise mechanism for the emergence of new miRNAs and their specific target interaction (Fig. 2A):

1. *Inverted duplication*: An inverted duplication event of a complete or partial founder gene, which could be a protein-coding gene or pseudogene (Kasschau et al., 2007; Guo et al., 2009), results in the formation of palindromic sequences arranged in a head-to-head or tail-to-tail orientation.
2. *Transcription and siRNA production:* The locus harboring the inverted sequences is transcribed under the promoter of the founder gene, generating a long transcript capable of folding into perfectly complementary stem-loop structures. This structure is then processed by DCL3/4, producing multiple siRNA duplexes that could regulate the founder gene and its family members through RNA interference at post-transcriptional or chromatin levels.
3. *Mutation and miRNA Formation*: Over time, sequence variation at the inverted duplication locus leads to the formation of imperfect stem-loop structures, allowing for processing by DCL1 instead of DCL3/4. This transition marks the emergence of miRNAs.
4. *Functional targeting*: The nascent miRNAs regulate the founder gene or its family members by pairing with homologous sequences. Through subsequent duplication events and the retention or loss of sequences complementarity, novel regulatory networks may develop.
5. *Further divergence*: As the miRNA sequences continue to diverge, constrained by the requirements for stem-loop structure and DCL1 recognition, only the mature miRNA and miRNA* sequences retain resemblance to the original founder gene or closely related family members.

Substantial evidence supports this hypothesis across both plants and animals (Voinnet, 2009; Cui et al., 2017; Baldrich et al., 2018; Fridrich et al., 2020; Wang et al., 2024). For instance, miR482/miR2118 in spruce emerged from inverted duplications of *Nucleotide-binding site Leucine-rich repeat* (*NBS-LRR*) genes (Xia et al., 2015). Subsequent study revealed that *NBS-LRR* genes periodically give rise to new miRNAs via this mechanism, typically targeting conserved sequences within *NBS-LRR* genes (Zhang et al., 2016). In ancestral wheat, miRW1 originated from an inverted duplication of W1-COE and/or W2-COE, regulating the glaucous trait by repressing its founder genes (Huang et al., 2017). Similarly, miR5200 evolved through an inverted duplication of *FLOWERING LOCUS T* (*FT*) in the grass subfamily Pooideae, regulating target genes under short-day photoperiods (McKeown et al., 2017). Intriguingly, recent research found that miR858, which contains ultra-long stem-loop transcripts, likely originated from the *TT2-like* gene (*MYB*) in seed plants through inverted duplication (Wang et al., 2024). Furthermore, miR444 and miR824 independently arose from partial inverted duplications of antisense target genes, resulting in the natural antisense arrangement between the new *MIR* gene and its target gene (Gramzow et al., 2020). This finding significantly extends the original model of genic inverted duplication.

3.1.2 Derived from retrotransposons

Prior to the hypothesis of miRNA derived from TEs was introduced, the connection between siRNAs and TEs was already established in both plants and animals through mechanisms like post-transcriptional gene silencing and RNA interfering (Matzke et al., 2001a, 2001b). Llave et al. (2002) identified miRNA-like sequences in *Arabidopsis thaliana* that were embedded within transposon-like sequences, leading to the conjecture that TEs or their derivatives could serve as sources of miRNAs (Mette et al., 2002).

Subsequently, Smalheiser & Torvik (2005) discovered that a dozen of *MIR* genes in mammals contained repeat sequences from Class I TEs, specifically retrotransposons such as long interspersed nuclear elements (LINEs), short interspersed nuclear elements (SINEs), and long terminal repeats (LTRs). They noted that miRNA stem-loop structures could be formed through the transposition and subsequent inverted tandem arrangements of LINE-2 segments. Some of these transcripts may eventually give rise to miRNAs (Jo et al., 2019; Lee et al., 2019; Guo et al., 2022b). The proposed evolutionary process includes the following steps (Fig. 2B):

1. *Inverted duplication or transposition*: Two consecutive retrotransposons, arranged in opposite configurations through inverted duplication or transposition into adjacent regions, may generate transcripts via readthrough events.
2. *siRNA production*: These transcripts could fold into long, perfect stem-loop structures, which trigger the production of siRNAs. These siRNAs specifically target and degrade transcripts derived from TEs, acting as a control mechanism to prevent TE mobilization (Castel & Martienssen, 2013).
3. *miRNA formation*: Over time, sequence diversification at the palindromic region leads to the formation of imperfect stem-loop structures, ultimately resulting in

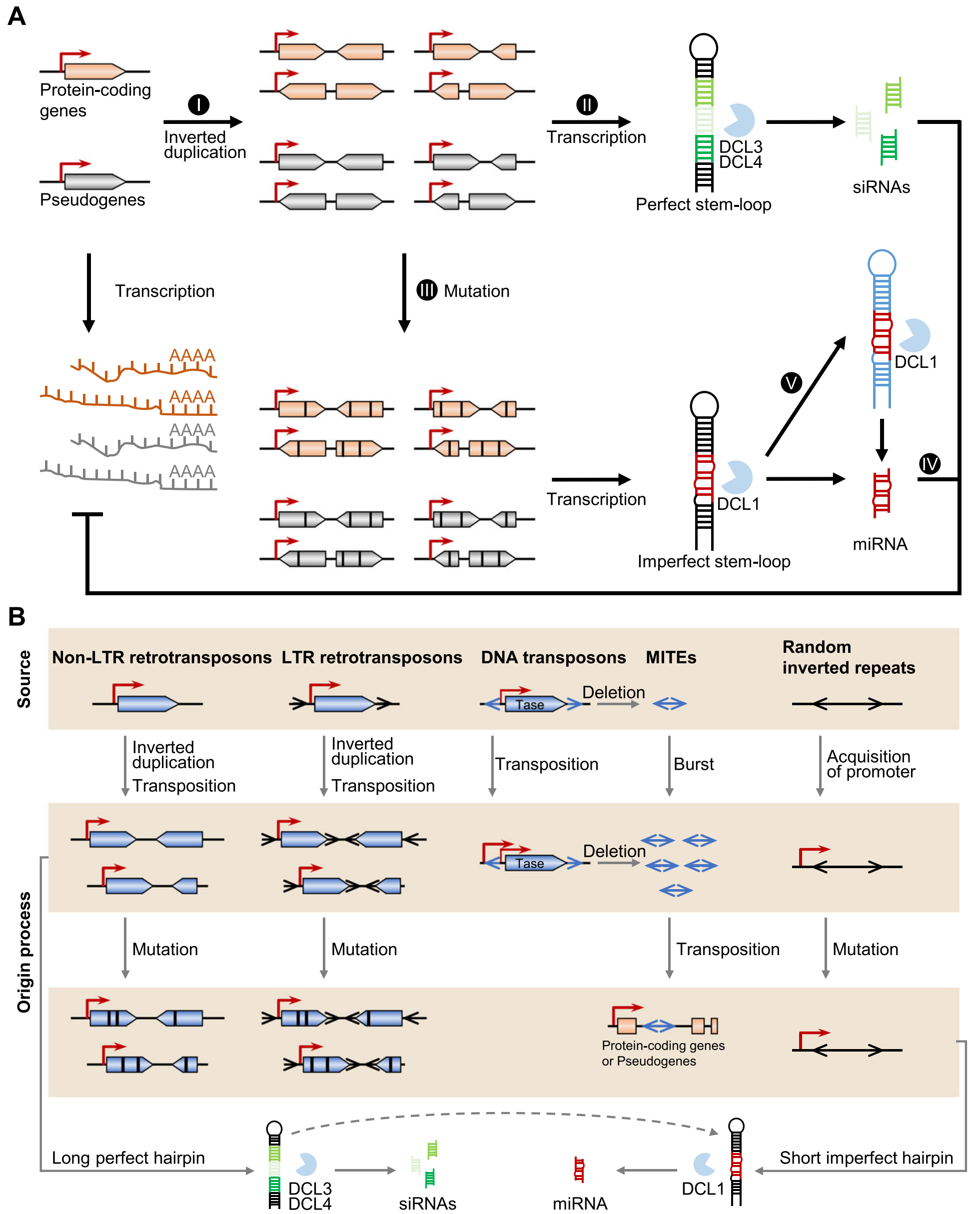

Fig. 2. Summary of models for the *de novo* origination of miRNAs in plants. **A,** miRNAs origination via inverted duplication of target genes. The process of is outlined in five steps: I. inverted duplication, II. transcription and siRNA production, III. mutation and miRNA formation, IV. functional targeting, V. further divergence. Detailed descriptions of each step are provided in Section 3.1.1. **B,** miRNAs origin from retrotransposons, DNA transposons and spontaneous evolution. The red arrows indicate acquisition of transcriptional activity. Dark arrows at LTR retrotransposons indicate long terminal repeats. Blue arrows at DNA transposons and MITEs indicate terminal inverted repeats. Tase, transposase. Black vertical lines on the exons indicate sequence mutations.

the generation of miRNAs. However, the mechanism by which these miRNAs acquire their target genes remains controversial.

3.1.3 Derived from DNA transposons

Although a mammalian miRNA (miR340) that shared sequences with class II elements, also known as DNA transposons, was already identified in Smalheiser and Torvik's study (Smalheiser & Torvik, 2005), it was only later that the broader recognition of miRNA origins from DNA-type elements emerged following sequence comparative sequence analyses in the human genome (Piriyapongsa & Jordan, 2007; Piriyapongsa et al., 2007). These studies demonstrated that certain human *MIR* genes were derived from MITEs, a special category of non-autonomous DNA transposons. Subsequent research in *Arabidopsis* and rice led to the proposal of a model explaining the evolutionary transition from DNA transposon-based siRNAs to MITE-based miRNAs (Piriyapongsa & Jordan, 2008; Zhang et al., 2011).

This model suggested that siRNAs were initially derived from the two terminal inverted repeats (TIRs) of full-length DNA-type elements, which folded into stem-loop structures with perfectly complementary bases. Over time, these full-length elements lost their coding sequences of transposases, resulting in the formation of MITEs, which retained only the TIRs and two target site duplications (TSDs). The amplification of MITEs led to their proliferation within the genome. The secondary structures encoded by MITEs, characterized by nearly perfect hairpins, were then targeted for cleavage by DCL1, leading to the production of mature miRNAs.

Recently, our team elucidated a transposition–transcription mechanism through which MITEs are converted into young miRNAs, thereby shaping the miRNA repertoire in angiosperms (Guo et al., 2022b). This process involves several steps (Fig. 2B):

1. *MITE amplification*: MITEs, particularly from the *Mutator*, *Tc1/Mariner*, and *PIF/Harbinger* superfamilies, undergo amplification in the genomes of angiosperms.
2. *Transposition into introns*: The rapid proliferation of MITEs significantly increases the likelihood of MITEs insertion into intronic regions.
3. *MITE transcription*: MITEs are transcribed into RNAs by hitchhiking the transcriptional machinery of their host gene.
4. *miRNA formation*: The transcripts fold into hairpin structures relying on the TIR-seeded regions of MITEs, which are then recognized by DCL1 complex, generating new miRNAs.
5. Functional targeting: MITE-derived miRNAs preferentially regulate genes involved in responses to environmental stimuli, potentially enhancing plant adaptation to diverse environments.
6. Collateral selection: In some cases, the selection of a miRNA may also result in the collateral selection of the host gene as a sole miRNA precursor through pseudogenization of host gene.

3.1.4 Spontaneous evolution

The hypothesis of spontaneous evolution suggests that miRNAs could originate randomly from the numerous inverted repeats scattered throughout the genome. This idea was first proposed by Svoboda and Di Cara (2006) in their study of animal miRNAs, where they introduced the random selection hypothesis. They posited that putative miRNAs could be generated from random inverted repeats that serve as substrates for Drosha and Dicer proteins. Given the relatively low sequence complementarity required for miRNA targeting in animals, these putative miRNAs could randomly acquire targets, creating a reservoir from which functional miRNAs could emerge.

This concept was later extended to plants considering the established identification of vast numbers of imperfect inverted repeats in the *Arabidopsis* genome (Jones-Rhoades & Bartel, 2004). Axtell (2008) speculated, and Felippes et al. (2008) later confirmed, that these random sequences could give rise to miRNAs in plants. According to this model, the inverted repeats dispersed throughout the genome have the potential to fold into hairpin structures when transcribed into RNA (Fig. 2B). Given the abundance of these inverted repeats relative to the number of miRNAs, it is plausible that some of these foldbacks could acquire promoters, enabling their transcription. Once transcribed, these foldbacks could become substrates for DCL1, forming new miRNAs. For instance, in *Arabidopsis*, a candidate pre-miRNA foldback known as mpss05 was found to align with two regions that may originated from the duplication of a chromosomal fragment (Felippes et al., 2008). This idea is reinforced by the observation that the precursor arms of many young *MIR* genes do not exhibit sequence similarity to any other sequences in the genome (Chen, 2009).

While the spontaneous evolution model offers a plausible mechanism for the random generation of miRNAs from inverted repeats, it faces several limitations from our empirical perspective. One of the primary challenges is the reliance on the random acquisition of functional promoters by these inverted repeats. The probability that these sequences, which are initially non-coding and lack regulatory elements, will acquire the necessary promoter activity to be transcribed into RNA is relatively low. Additionally, the model does not sufficiently explain how these randomly originated miRNAs would consistently acquire target mRNAs with the necessary sequence complementarity. These issues raise questions about the efficiency and evolutionary feasibility of the spontaneous evolution model in explaining the origin of functional miRNAs.

3.1.5 Origination by template switching mutations

Recent studies introduced the concept of miRNA origination through template switching during DNA replication as a novel mechanism for the generation of *de novo* miRNAs, particularly with the primate lineage (Mönttinen et al., 2023). Template switching is a mutation mechanism associated with DNA replication, capable of introducing complex genetic changes in a single event, such as the creation of perfect base-pairing hairpin structures (Löytynoja & Goldman, 2017; Walker et al., 2021; Löytynoja, 2022). Initially, this mechanism was proposed to explain the conversion of quasi-palindromic sequences into perfect palindromes during replication (Ripley, 1982; Dutra & Lovett, 2006). Template switching could occur within the same strand (intrastrand switch), leading to loopback hairpin formation, or by briefly jumping to the opposing strand (interstrand switch), before resuming replication on the original strand. This process results in the one-step formation of inverted repeats, which are key for miRNA formation.

In an earlier study, Mönttinen et al. demonstrated that template switching not only contributes to the generation of new RNA hairpins but also helps maintain these complementary structures over time (Mönttinen & Löytynoja, 2022). These hairpins, formed during the replication process, could serve as substrates for miRNA biogenesis machinery, thereby providing a potential pathway for the formation of miRNA stem-loop structures without the need for pre-existing inverted repeats. Subsequent studies reconstructed the ancestral histories of *MIR* genes and identified mutation patterns consistent with the template switching (Mönttinen et al., 2023). Their findings revealed that template switching mutations contributed to the emergence of over 6000 hairpin structures in the primate lineage, leading to the creation of at least 18 new human *MIR* genes, accounting for

26% of the miRNAs that originated in primates. Although this mechanism appears random, the miRNAs generated by template switching often enriched in introns, enabling their co-expressed with host genes.

It is important to note that this mechanism has not yet been demonstrated in plants. However, the hypothesis broadens our understanding of miRNA origination, suggesting that miRNAs may arise from inherent DNA replication rather than relying on specific genomic elements. Exploring this model in plants is a worthwhile direction for future research.

3.2 Expansion through duplication

The duplication of *MIR* genes is a crucial mechanism that drives the expansion of miRNA families within genomes. Although plants have fewer miRNA families than animals, their families are larger, with multiple members arising from duplication events (Jones-Rhoades et al., 2006; Li & Mao, 2007). These events, analogous to the well-studied duplications of protein-coding genes, include whole-genome duplications (WGDs), segmental duplications (SDs), and tandem duplications (TDs) (Ohno, 1970; Zhang, 2003; Li & Mao, 2007). These processes enable substantial expansion of *MIR* genes, adding complexity to transcriptional regulatory networks.

WGDs are particularly pivotal in the expansion of miRNA families in plants, leading to a significant increase of the miRNA repertoire. This impact is evident when comparing closely related species that diverged before and after a WGD event (Cui et al., 2017). For instance, in the Fabaceae family, *Glycine max* has 2,787 annotated *MIR* genes (Liu et al., 2016), while *Phaseolus vulgaris* has only 216 (Nithin et al., 2015). Similarly, within the Brassicaceae family, *Brassica rapa* possesses 680 *MIR* genes compared to just 80 in *Capsella rubella* (Smith et al., 2015; Sun et al., 2015). Although many miRNAs derived from WGDs may undergo rapid degradation (Zhao et al., 2015), the overall effect of WGDs is a pronounced and swift expansion of miRNA families.

In addition to WGDs, SDs and TDs further contribute to the expansion of miRNA families by duplicating specific genomic regions or individual *MIR* genes (Sun et al., 2012; Hajieghrari Behzad, 2020). In *A. thaliana*, at least 17 miRNA families have expanded through SDs and six through TDs (Maher et al., 2006). For example, miR156b/miR156c and miR156e/miR156f were identified as TDs within the same intergenic region. While SDs and TDs could lead to the formation of miRNA clusters, especially in animal genomes, *MIR* genes in plants from the same family are frequently dispersed throughout the genome (Wang et al., 2016, 2023; Singh et al., 2020). This widespread distribution suggests that plant genomes have undergone extensive shuffling since the amplification of these miRNA families. However, a notable exception to this pattern is the miR395 family, which remains tightly clustered in several plant genomes and could be transcribed as a single transcript (Guddeti et al., 2005; Liang et al., 2010; Singh et al., 2020).

Following duplication, miRNA paralogs often retain identical sequences throughout the entire *MIR* gene (Li & Mao, 2007). Over time, these paralogs may accumulate mutations leading to functional divergence, including subfunctionalization, where the original function is partitioned between the duplicates, or neofunctionalization, where a duplicate acquires a new function (Zhao et al., 2015; Baldrich et al., 2018). For example, members of miR172 family in *Arabidopsis* demonstrate both overlapping and specialized functions (Lian et al., 2021; Ó'Maoiléidigh et al., 2021). Under long-day conditions, miR172a and miR172b predominantly promote flowering, with high expression in the leaf vasculature. In contrast, miR172d serves as the primary contributor to flowering under short-day conditions, with high expression in the shoot apical meristem. Given the sequence similarity observed across different miRNA families, such as miR156/miR157/miR529, or miR159/miR319, it is plausible that novel *MIR* genes could arise through duplication followed by subsequent mutation (Palatnik et al., 2007; Nozawa et al., 2012; Morea et al., 2016).

Moreover, miRNA duplication contributes to the robustness and flexibility of regulatory networks. By increasing the dosage of mature miRNAs, duplication improves the efficiency of target gene silencing (Abrouk et al., 2012; Shi et al., 2022). The presence of multiple paralogous miRNAs with identical or similar mature sequences also provides a buffer against mutations, ensuring that essential regulatory functions are preserved even as individual miRNA copies diverge.

3.3 An open question regarding miRNA loss

The turnover of miRNAs is a dynamic process involving both the rapid gain and loss of these regulatory elements. Extensive research on miRNA loss has been conducted across various animal lineages. For instance, Ma et al. (2021) conducted a comprehensive analysis of miRNA gain and loss across 152 arthropod species, revealing a significant loss of miRNA families in Paraneopteran, closely associated with morphological simplification. In *Drosophila*, comparative analysis showed a miRNA loss rate of approximately 11.7 miRNAs per million years (Lu et al., 2008). Further studies indicated that 87% of miRNAs are lost within 4–30 million years after their emergence, with only 6% persisting beyond 60 million years (Lyu et al., 2014). Additionally, Zhang et al. (2008) identified a tandem miRNA family with seven members on chromosome 19 across six primate species that were lost due to Alu element insertions, deletions, and mutations. In contrast, research on miRNA loss in plants is relatively limited. Early work by Fahlgren et al. (2010) attempted to estimate miRNA loss by comparing orthologous miRNAs between *A. thaliana* and *Arabidopsis lyrate*. They found that 10 miRNAs unique to *A. thaliana* also lacked an ortholog in *Capsella rubella*, as did nine miRNAs specific to *A. lyrate*.

While the generation of new miRNAs has been extensively studied, the phenomenon, mechanisms and evolutionary significance of miRNA loss, particularly in plants, remain poorly understood. Herein, we propose three interconnected steps to advance the study of miRNA loss.

3.3.1 Identification of miRNA loss events across broad lineages

Accurately and comprehensively identifying miRNA loss events is fundamental for subsequent analyses. With the growing availability of genomic and sRNA-seq resource, it has become feasible to annotate miRNAs across a wide range of plant lineages. Therefore, miRNA loss events could be detected through comparing the present or absent among distinct species. Given the uneven sampling and varying evolutionary timescales at different taxon, a multi-dimensional approach to analyses miRNA loss patterns, such as within a specific family, genus or species, may yield more robust insights.

Due to the frequent loss of miRNAs, focusing on intra-genus or intra-species levels may be particularly effective in uncovering the mechanisms driving miRNA loss. Genera such as *Arabidopsis*, *Solanum*, and *Oryza*, already possess numerous high-quality genomes and sRNA-seq datasets, making them ideal candidates for targeted investigations. Additionally, the availability of over 900 pan-genome projects (Xie et al., 2024) presents a promising path to explore miRNA loss within populations of individual species, potentially offering deeper understanding of miRNA dynamics.

3.3.2 Exploration of mechanisms underlying miRNA loss

The loss of miRNAs may arise from mutations that gradually disrupt miRNA biogenesis. Mutations, including base substitutions, deletions, and insertions, can lead to sequence erosion over time, ultimately rendering miRNA loci non-functional. Evaluating the types and frequencies of these mutations is critical for understanding their contributions to miRNA loss.

Abrupt disruptions to miRNA biogenesis are another potential contributor to miRNA loss. One driver of such event is transposition, particularly involving MITEs. Previous research identifies MITEs as a major source of miRNAs in angiosperms (Guo et al., 2022b). However, young MITE-derived miRNAs may relocate to transcriptionally inactive genomic regions, leading to miRNA functional loss. Additionally, chromosomal rearrangements, such as unequal crossing-over during meiosis, can physical removal or relocation of miRNA loci.

3.3.3 Evolutionary significance of rapid miRNA loss

The loss of miRNAs may reshape the architecture of gene regulatory networks by altering the interactions among transcription factors, miRNAs, and target genes. By constructing regulatory networks of these elements, the roles of lost miRNAs can be mapped, enabling comparisons of network structures before and after their loss. Such analyses provide valuable insights into how the absence of specific miRNAs affect the overall complexity and functionality of regulatory networks.

In addition, miRNA loss can influence gene functions and plant traits through the deregulation of target genes. Functional annotations of these targets may shed light on the consequences of miRNA loss, and help elucidate its impact on plant adaptability.

Furthermore, the sequences of lost miRNAs may acquire new regulatory roles, contributing to novel functionalities. We hypothesize two scenarios for such functional repurposing. First, the sequence of a lost miRNA may evolve into a new miRNA family, regulating distinct target genes and performing new functions. Second, the original miRNA sequence could transform into other types of regulatory molecules, such as siRNAs or non-autonomous DNA transposons, adopting alternative roles within the genome.

4 Conclusion

The study of miRNAs in plants reveals a dynamic and complex landscape during evolutionary trajectory. Despite their rapid turnover, miRNAs play essential roles in key biological processes, creating a paradox between their transient nature and functional importance. The existence of conserved miRNA families across land plants, alongside more transient lineage-specific and species-specific miRNAs, illustrates the delicate balance between evolutionary innovation and the conservation of critical regulatory functions.

The *de novo* origination of miRNAs entails overcoming three significant challenges, appropriate transcriptional activation, structural requirements for biogenesis, and functional integration into gene regulatory networks. These processes differ markedly between plants and animals, reflecting the distinct evolutionary paths of miRNAs among these organisms (Axtell et al., 2011; Moran et al., 2017). Various mechanisms have been proposed to explain miRNAs origination in plants, including inverted duplication of target genes, derivation from retrotransposons or DNA transposons, and spontaneous evolution. However, a systematic analysis reveals that these known models account for only 29.6% of miRNAs origination in angiosperms, leaving a substantial number of new miRNAs unexplained. This suggests the existence of yet unidentified mechanisms, which may be crucial for understanding how plants frequently generate new miRNAs.

Duplication events, such as whole-genome duplications, SDs, and TDs, have significantly contributed to the expansion of miRNA families in plants. These events promote the functional diversification of miRNA paralogs, enhancing the robustness and flexibility of gene regulatory networks (Cui et al., 2017; Baldrich et al., 2018).

The phenomenon of miRNA loss remains an underexplored frontier in miRNA evolution research, particularly in plants. Given its pivotal role in the rapid turnover of miRNAs, understanding the mechanisms behind miRNA loss is crucial for decoding the dynamic nature of miRNAs. miRNA loss has the potential to reshape gene regulatory networks, modify gene functions, and influence plant traits. Exploring the molecular drivers of miRNA loss and their evolutionary significance, will not only enhance our understanding of the mechanisms shaping plant genomes but also provide deeper insights into how plants adapt to environmental challenges and thrive across diverse ecological niches.

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Conflicts of Interest

The authors declare no competing financial interest.

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